COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CpE21S2
DATE PERFORMED:	7/14/2026	**DATE SUBMITTED:**	7/15/2026
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Hands-on Activity 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

ILO A

Create a function that will take two numbers and display the sum.

```cpp
//Fernandez, Cedric S. CpE21S2
#include <iostream>

//This is the function prototype which is a semi headsup to the compiler so it doens't result in an error
void twoNum(int num1, int num2);

//The section where the compiler start and calls the function twoNum
int main () {
    twoNum(5, 10);
    return 0;
}

//After main calls for the function it goes to this section and does the computation then after showing the sum it goes
//back to main and ends the program
void twoNum(int num1, int num2) {
    int sum = num1 + num2;
    std::cout << "The sum of the two numbers is: " << sum;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

[Done] exited with code=1 in 0.4 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_A.cpp -o ILO_A && "c:\Users\csf18\Desktop\Codes\"ILO_A
ILO_A.cpp: In function 'void twoNum(int, int)':
ILO_A.cpp:14:55: error: 'num' was not declared in this scope; did you mean 'sum'?
   14 |     std::cout << "The sum of the two numbers is: " << num;
      |                                                       ^~~
      |                                                       sum

[Done] exited with code=1 in 0.437 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_A.cpp -o ILO_A && "c:\Users\csf18\Desktop\Codes\"ILO_A
ILO_A.cpp: In function 'void twoNum(int, int)':
ILO_A.cpp:14:55: error: 'num' was not declared in this scope; did you mean 'sum'?
   14 |     std::cout << "The sum of the two numbers is: " << num;
      |                                                       ^~~
      |                                                       sum

[Done] exited with code=1 in 0.337 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_A.cpp -o ILO_A && "c:\Users\csf18\Desktop\Codes\"ILO_A
The sum of the two numbers is: 15
[Done] exited with code=0 in 0.836 seconds
```

Create a function that will return whether variable A is greater than variable B.

```cpp
//Fernandez, Cedric S. CpE21S2
#include <iostream>

//serves as a headsup for our compiler to avoid error
bool greaterVal(int VarA, int VarB);

//where the compiler start and calls the function greaterVal
int main () {
    greaterVal(5, 10);

//Makes it so that our output is either true or false and not 1 or 0
    std::cout << std::boolalpha;

//Displays the value whether A is greater than B then after ends program
    std::cout << "Variable A is greater than Variable B: " << greaterVal;
    return 0;
}

//where the compiler skips off to in order to return the value of greaterVal
bool greaterVal(int VarA, int VarB) {
    if (VarA > VarB) {
        return true;
    }
    else {
        return false;
    }
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_Test.cpp -o ILO_Test && "c:\Users\csf18\Desktop\Codes\"ILO_Test
ILO_Test.cpp: In function 'bool greaterVal(int, int)':
ILO_Test.cpp:18:1: warning: no return statement in function returning non-void [-Wreturn-type]
   18 | }
      | ^

[Done] exited with code=3221225501 in 1.122 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_Test.cpp -o ILO_Test && "c:\Users\csf18\Desktop\Codes\"ILO_Test
Variable A is greater than Variable B: 1
[Done] exited with code=0 in 0.962 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_Test.cpp -o ILO_Test && "c:\Users\csf18\Desktop\Codes\"ILO_Test
Variable A is greater than Variable B: true
[Done] exited with code=0 in 1.028 seconds
```

Create a function that will take two Boolean values and display the result of all logical operations

then return true if it was a success.

```cpp
// Cedric Fernandez CPE21S2
#include <iostream>

//Prototype function that acts as a heads up to avoid errors in execution in the code
bool trueOrfalse(bool x, bool y);

//Calls the prototype function
int main() {
    trueOrfalse(true, false);
    return 0;
}

//executes the prototype function trueOrfalse
bool trueOrfalse(bool x, bool y) {
    // Displays true or flase instead of 1 and 0
    std::cout << std::boolalpha;
    // Displays the AND, OR, and NOT result
    std::cout << "AND result: " << (x && y) << std::endl;
    std::cout << "OR result: " << (x || y) << std::endl;
    std::cout << "NOT x result: " << (!x) << std::endl;

    //Return true to show success operation
    return true;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
[Done] exited with code=1 in 2.099 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_Test.cpp -o ILO_Test && "c:\Users\csf1
C:/msys64/ucrt64/bin/../lib/gcc/x86_64-w64-mingw32/15.2.0/../../../../x86_64-w64-mingw32/bin/l
`trueOrfalse(bool, bool)'
collect2.exe: error: ld returned 1 exit status

[Done] exited with code=1 in 0.515 seconds

[Running] cd "c:\Users\csf18\Desktop\Codes\" && g++ ILO_Test.cpp -o ILO_Test && "c:\Users\csf1
AND result: false
OR result: true
NOT x result: false

[Done] exited with code=0 in 1.632 seconds
```

ILO B

```cpp
// Cedric Fernandez CPE21S2
#include <iostream>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}

int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
[Running] cd "c:\Users\cs[illegible]\Desktop\Codes\" && g++ ILO_Test.cpp -o
The shape is a valid triangle.
```

Sections	Answer / Code Elements
Header File Declaration Section	#include <iostream>
Global Declaration Section	*None*
Class Declaration	class Triangle { ... };
Main Function	int main() { ... }
Method Definition	Triangle::Triangle(...) Triangle::setAngles(...) Triangle::validateTriangle(...)

Observations and comments.

One thing that I noticed during following these steps is the line of return (totalAngle <= 180); which showed that it doesn't follow the rule of a triangle interior angle strictly needs to be 180 if it isn't exactly 180 then it won't be classified as a triangle

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a C++ program to swap the two numbers in different variables.
2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
3. Create a C++ program that has a function that will calculate the distance between two points.

FULL CODE

```
//Cedric S Fernandez CpE21S2
#include <iostream>
//Enables us to use adavance math like square root and power
#include <cmath>

//decleration of prototype function we used void here because we plan to also make each said function to output text
//along with the answer to their final computation or output
void swapVar ();
void kelvinTofahrenheit ();
void distanceCalculator();

int main () {

int choice;

//Displays the menu and ask for users choice on what option they want
std:: cout<< " Welcome! please choose one of the following" << std::endl;
std:: cout<< "1.) Swap values between 2 variables" << std::endl;
std:: cout<< "2.) Convert kelvin to Fahrenheit" << std::endl;
std:: cout<< "3.) Calculate distance between 2 points" << std::endl;
std:: cout<< "Please input one of the following from the menu (1-3): ";
std::cin >> choice;

//Is the portion that makes the menu section work with the use of switch cases and prototype functions
switch (choice) {
    case 1:
    swapVar();
    break;

    case 2:
    kelvinTofahrenheit();
    break;

    case 3:
    distanceCalculator();
    break;

    default:
    std::cout << "Invalid inpute please choose from 1-3: ";
    break;
    }
    return 0;
}
```

```cpp
//Section where the prototype function are stored and executed when called

//Variable swapper
void swapVar() {
int num1, num2;
std::cout << "Enter first number (Num1): ";
std::cin >> num1;
std::cout << "Enter second number (Num2): ";
std::cin >> num2;

std::cout << "Before Swap Var A = " << num1 << ", Var B = " << num2 << std::endl;
std::swap(num1, num2);
std::cout << "After Swap Var A = " << num1 << ", Var B = " << num2 << std::endl;
}

//Kelvin to fahrenheit converter
void kelvinTofahrenheit() {
double kelvin, fahrenheit;
std::cout << "Enter temperature in Kelvin: ";
std::cin >> kelvin;

//Just used the formula for Kelvin to fahrenheit online: F = (K - 273.15) * 9/5 + 32
fahrenheit = (kelvin - 273.15) * 9.0 / 5.0 + 32.0;

std::cout << kelvin << "K is equal to " << fahrenheit << " F ";
}

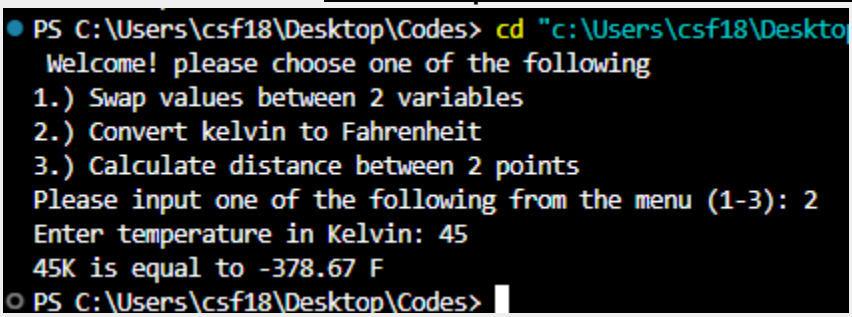
// Distance calculator
void distanceCalculator() {
double x1, y1, x2, y2, distance;
std::cout << "Enter Coordinates for Point 1 (x1 y1): ";
std::cin >> x1 >> y1;
std::cout << "Enter Coordinates for Point 2 (x2 y2): ";
std::cin >> x2 >> y2;

// Just used Euclidean Distance Formula here from online which is sqrt((x2-x1)^2 + (y2-y1)^2)
distance = std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
std::cout << "The distance between the two points is: " << distance << std::endl;
}
```

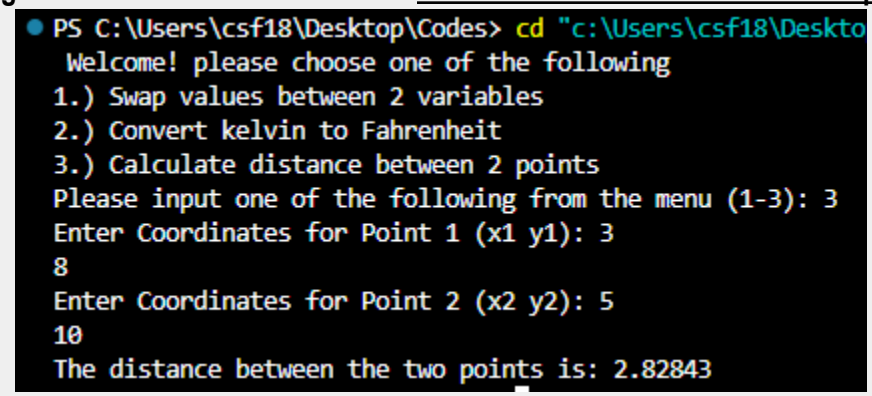
Output **for program to swap the two numbers in different variables.**

```
PS C:\Users\csf18\Desktop\Codes> cd "c:\Users\csf18\Desktop
 Welcome! please choose one of the following
1.) Swap values between 2 variables
2.) Convert kelvin to Fahrenheit
3.) Calculate distance between 2 points
Please input one of the following from the menu (1-3): 1
Enter first number (Num1): 5
Enter second number (Num2): 10
Before Swap Var A = 5, Var B = 10
After Swap Var A = 10, Var B = 5
```

Output **for program that has a function to convert temperature in Kelvin to Fahrenheit.**



Output **for program that has a function that will calculate the distance between two points.**



4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
// Cedric S Fernandez CpE21S2
#include <iostream>
// Needed for squareroot and power in solving
#include <cmath>


class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;
    //We added this section so that the program can start accepting side length so we can get the parameter and area of a triangle
    double sideA, sideB, sideC;

public:
    // Constructor declaration
    Triangle(double A, double B, double C, double sA, double sB, double sC);

    void setAngles(double A, double B, double C);
    const bool validateTriangle();

    //introduced 3 new prototype functions so that we won't get an error when calling for them
    double calculateArea();
    double calculatePerimeter();
    std::string determineType();
};

int main() {
    // Create a triangle with angles (40, 30, 110) and sides (7.0, 5.5, 10.3)
    Triangle set1(40, 30, 110, 7.0, 5.5, 10.3);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
        std::cout << "Perimeter: " << set1.calculatePerimeter() << " units\n";
        std::cout << "Area: " << set1.calculateArea() << " sq. units\n";
        std::cout << "Triangle Classification: " << set1.determineType() << "\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

```cpp
// Fixed Constructor: Now saves BOTH angles AND sides to our private variables
Triangle::Triangle(double A, double B, double C, double sA, double sB, double sC) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;

    //The added section where we'll store the side length
    sideA = sA;
    sideB = sB;
    sideC = sC;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}

//Perimeter calculator
double Triangle::calculatePerimeter() {
    return sideA + sideB + sideC;
}

// section that finds the area using Heron's Formula found online
double Triangle::calculateArea() {
    double s = calculatePerimeter() / 2.0;
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

//Triangle type identifier
std::string Triangle::determineType() {
    if (!validateTriangle()) {
        return "Invalid Triangle";
    }
```

```
    // Checks if any single angle is obtuse (greater than 90)
    if (angleA > 90 || angleB > 90 || angleC > 90) {
        return "Obtuse-angled";
    }
    // Checks if all angles are acute (less than 90)
    else if (angleA < 90 && angleB < 90 && angleC < 90) {
        return "Acute-angled";
    }
    // Any other case (like having exactly a 90 degree angle)
    else {
        return "Others (Right-angled)";
    }
}
```

```
The shape is a valid triangle.
Perimeter: 22.8 units
Area: 18.0427 sq. units
Triangle Classification: Obtuse-angled
```

Conclusion

Through this activity I gained a deeper insight on the proper use of prototype functions and specifically during the procedure where I had to know how to properly arrange the boundaries between the driver main() and the worker functions where the prototype function are basically being executed, in addition to this during the supplementary activity while I had an easy time on the first few section around the swapping variable values, temperature conversion and finding the distance since it was mostly trying to learn and recall function that'll help execute my code like using <cmath> so that the compiler would be able to handle things like square root and power or even just using the swap function in order to swap value between two variables since it mostly came to learning those said function or how they are structured to actually make them work the last few parts though is the section I had most difficulties in mostly because it involved working with an already made code that in some parts I had no clue what it did so occasionally when i tried to add or even remove something to that specific line of code sometimes it'd just mess up the overall code and it won't be abke to run so I'd have to start all over again or just try to nitpick every line of code just to see what I did wrong that spent too much time to do. Especially when it came to applying on solving the herons formula and also the triangle classification section cause that was the section I got stuck on because I didn't know what I did wrong.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concept**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Had to ask for examples on how to use a specific function or how structure a prototype function properly so that it can work properly as intended**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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